Theories, Models, Subject Matters, and Questions

Felipe Morales Carbonell

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Outline

1) Subject matter in theories and models
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   - What are models about?
2) Intended models
3) Intentionalism vs externalism concerning subject matters
4) Aboutness and denotation
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5) An aboutness-baseed alternative to DEKI
6) Models as inquisitive devices

A peculiarity in Newton's *Principia*

What are the *Principia* about?

The traditional answer: *rational mechanics*. But what is that? 'The science, expressed in exact propositions and demonstrations, of the motions that result from any forces whatever and of the forces that are required for any motions whatever' (Preface).

> [...] the description of straight lines and circles, which is the foundation of geometry, appertains to mechanics. Geometry does not teach how to describe these straight lines and circles, but postulates such a description. For geometry postulates that a beginner has learned to describe lines and circles exactly before he approaches the threshold of geometry, and then it teaches how problems are solved by these operations. To describe straight lines and to describe circles are problems, but not problems in geometry. Geometry postulates the solution of these problems from mechanics and teaches the use of the problems thus solved. [...] Therefore geometry is founded on mechanical practice and is nothing other than that part of universal mechanics which reduces the art of measuring to exact propositions and demonstrations.
>
> — PM, Preface, 381-382

Geometry is a part of mechanics! Geometry is a subject matter for mechanics because geometry deals with mechanical *problems*; rational mechanics deals with a subset of the problems of universal mechanics; and so on.

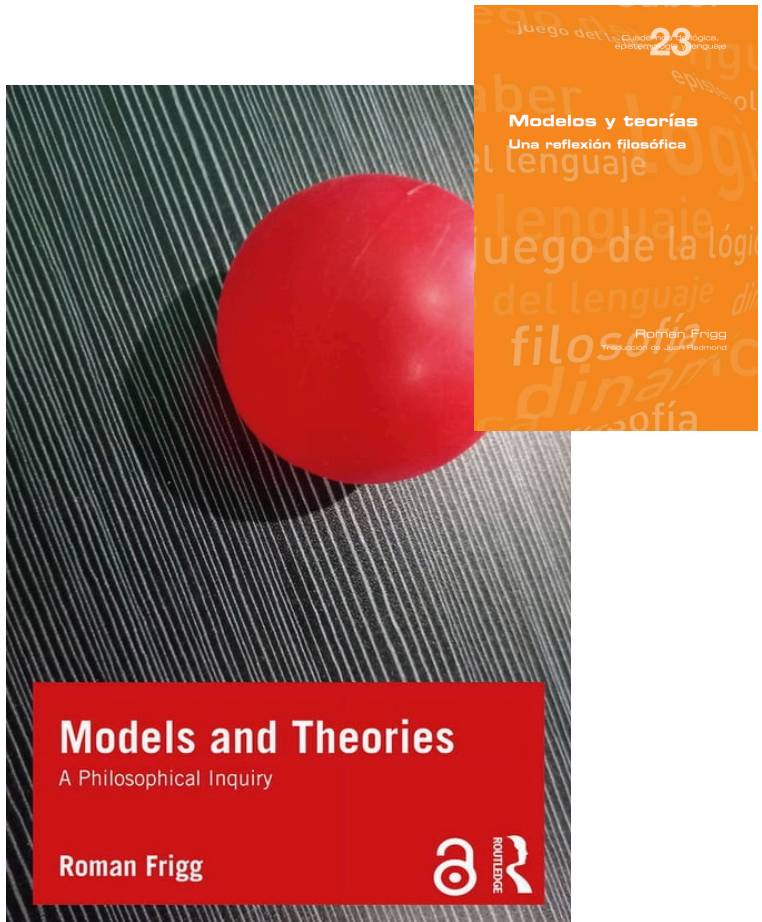
The idea

My proposal will be that theories and models get their subject matters by providing answer to questions.

Their subject matter is not what they *say*, but what they *answer*.

To illustrate the idea, I want to discuss how the notion of subject matter works in Roman Frigg's recent book, *Models and Theories* (MT).

I pick this work because it offers a systematic view of a popular way to understand models and their connection to theories (as its title promises).



Subject matter

- Subject matter is a separate dimension to meaning (Yablo, Berto, Fine…)
- A semantic object's subject matter is what it is about (in context*):
  1) 'Fido is happy' is about Fido, Fido's happiness, how Fido is, etc.,
  2) 'Bring me the ladder' is about the addressee performing an action,
  3) '✈️' is about flying,
- Different kinds of objects will give rise to different aboutness relations.
- Subject matters are linked to questions: 'Fido is happy' is linked to the question 'how is Fido feeling?'.
- There is a question that (indicative) sentences exactly answer: that is their exact subject matter.
- Depending on how we spell out theories and models, they will also have subject matters.

Subject matter in MT

a) MT's representationalism about theories endorses the claim that theories have worldly subject matters: *"A scientific theory is about something in the world"* (p. 157)

b) Models represent something beyond themselves: their *target systems*. They are what models are about.

   Target systems are not *phenomena* (Elliot-Graves 2020). Phenomena are the states and behaviors of real-world systems under study. Target systems are the aspects of the system that are studied in order to gain knowledge about phenomena. Target systems are close to *data*. But while theories are about phenomena, they are not about data.

c) The linguistic view of theories has it right, according to MT, that scientific theories describe their subject matters in formal languages. In the linguistic view, laws are formal sentences interpreted in terms of the theory's subject matter.

d) The distinction between language and metalanguage is presented in terms of theoretical subject matter. In this connection, subject matter is like reference.

e) A criticism that MT mounts against Suppes' semantic account is that it misplaces the subject matter of theories. If theories consisted of set-theoretical predicates, they would lack empirical content and fail to be about anything in the world. The data matching and morphism accounts are potentials solutions to this problem. Because theories represent phenomena and not data, data matching fails. Pure morphism fails, because theories then fail to be about specific subject matters due to concerns related to Newman's problem.

f) The problem of unintended targets and unintended models is a mismatch between theories and intended targets. First order theories can be said to be about any of their models, even if unintended. Second order to the rescue? Not without trade-offs.

g) The representational capacities of scientific representations enable surrogative reasoning about their subject matters (pp. 186–187).

h) Frigg critically discusses Levy's (2015) account of direct representation and modelling, which makes explicit use of Yablo's (2014) account of subject matter.

i) Subject matter reappears in the discussion of Hughes DDI account of representation, and then on the features of DEKI. Hughes takes the relation of theories and subject matters to be one of denotation. In DEKI, denotation is replaced with Z-representation.

Internalism and externalism about subject matter

How do theories get their subject matters? Two basic options:

1) *Internalistically*: an item is assigned a subject matter by the user. It is a function of the user's intentions.
2) *Externalistically*: an item acquires its subject matter in context, independently of the user's intentions.

I will use the expression *intentionalism* as more or less synonymous with 'internalism'.

We can combine these: there might be dual theories, where subject matter is always determined internalistically and externalistically, and pluralist theories, where it is sometimes fixed internalistically and sometimes fixed externalistically.

We can apply this to the case of theories and models.

Intentionalism about theories and models

Most usual approaches are internalistic: the picture is that when we inquire into something, we make the decision to target and this defines the overall subject matter of the representations we produce in the process of inquiring into these targets of inquiry.

Elliot-Graves (2020): the selection of target systems begins with the choice to target a phenomenon that is present in a given domain.

Weisberg (2013):

> Choosing a target is very much in line with choosing a subject matter. Two scientists studying the Adriatic Sea after World War I might have very different interests: one in predatory-prey relations, and one in the effect of surface temperature on algae blooms. The choice of one topic or the other will result in the generation of a different target system from the same phenomenon. The choices are free, but there seems to be no anarchy or arbitrariness in this difference. (pp. 91–92)

Frigg's position and the case of unintended models

Frigg draws from Bogen & Woodward's (1988) account. What we intend to do is extract from the data an account that explains the underlying phenomena, which are our real targets of inquiry.

In Frigg's position, the role of intentions in the fixation of subject matter is more limited.

We see this is his treatment of *unintended models*. The problem is usually stated as: if a theory had unintended models, that would somehow count against the theory, because it would fail to capture the intended subject matter and it only.

Do theories get their subject matter when they are assigned intended models?

But:

- First, not in every case it is possible to start with a well-defined class of intended models. In these cases, finding models constitutes a scientific achievement, even if surprising.
- Second, there might not be any models at all, and the whole task is to find out if there are any. In these cases it makes no sense to talk about intended models, and the absence of such models is not a strike against theories.

The issue is that the assumption mentioned above is faulty: in many cases, the subject matter of theories can only be defined as whatever matches a set of relevant conditions.

So Frigg moves towards a form of anti-intentionalism, or at the very least some form of pluralism.

The challenge of targetless models

Moving towards an account of the subject matter of scientific representations that is at least partially anti-intentionalist rises some worries.

If the subject matter of scientific representations is only partially determined by intentions, it can happen that some apparently targetless models (because they are not intended to have targets) are not really targetless (because despite being intended not to have targets, they do).

Conversely, intentionalism seemingly disallows apparently targetless models from having targets.

Frigg takes the existence of targetless models as a given and thinks that an account of scientific representation should explain how it is possible for models to be targetless. This is what he calls the *Targetless Models Condition* (p. 188), which he uses to evaluate different theories of scientific representation.

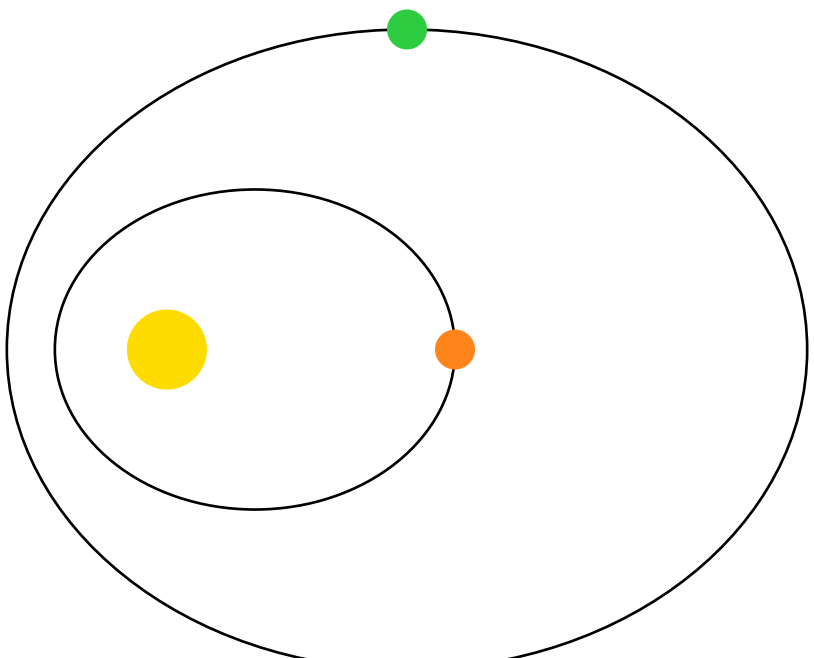
Levy (2015)

Levy (2015) proposes what can be called a *direct representation* account of models. The idea is that models do not have an independent existence (models systems are not objects standing in relations to their targets), but are merely a particular type of description of those targets that serves as props in certain games of make-believe that prescribe specific imaginings about the system.

A problem for views of this type: in appealing to the imagination, they lose epistemic friction. Imagining that something has certain properties is not sufficient to determine that whatever we imagine *has* those properties. Furthermore, it is not clear how whichever properties the imagines could be imagined to have could *actually* be had by the target. How to deal with this problem?

Levy (2015) appeals to yablo's (2014) notion of *partial truth*. Models are at least partially true of the targets that they represent, and this enables surrogative reasoning.

A model where Kepler’s first law holds is false of the solar system, but is partially true, as the Earth (at least) traces an orbit around the Sun that is approximately elliptic.



Levy on targetless models

Frigg notices (MT, p. 270) that in Levy's account there are no genuinely targetless models.

- Some models are contingently targetless because the target has not been determined yet—but once they are determined, the proposal applies without change.
- In other cases, while it might appear that there are no targets, there really is a generalized target for which the model is still partly true. This happens in so-called 'hub-and-spoke' models, which contain hubs which are models with a target, connected to models without targets.

  For example, there is the family of models of populations with n sexes. The model with 2 sexes is a hub model: it has a real target. The model with 4 sexes is not a hub model. But because it is suitably connected to the 2-sex model, it can be said to be about a generalised target, namely, population growth in general.
- In other cases, the absence of a target gives reason to reject the label of 'model'.

Frigg against Levy

It seems as if Levy trades the initially compelling lack of reference to abstracta of direct representations for the posit of new abstracta:

> Notions like "population growth in general" do not single out concrete things that can serve as the object of an act of pretence, and even if one were to generalise the notion of an object so that the imagination can be about generalisations of that kind, they raise the same ontological worries as [...] fictional characters. One cannot at one reap the ontological benefits of viewing modelling as imagining something about a concrete object, while at the same time allowing for targets that are not concrete objects. (p. 271)

A response on behalf of Levy: "population growth in general" is a pattern of cross-world variations, supervenient on the relevant *concrete* parts of worlds in logical space.

The cost is not as bad as Frigg makes it out to be (if we couldn't account for this kind of pattern, we couldn't account for modal facts either–in my opinion, a much worse cost). 🤷.

Goodman on denotation

Frigg's favored view of representation, the DEKI account, builds on Goodman's (1968) notion of representation-as, by the way of Hughes (1997) and Elgin (2010). The key notion in Goodman's theory of representation is that of *denotation*.

For a representation to represent, it does not have to denote anything, but it must be the sort of symbol that denotes; thus, an entirely fantastical tale represents because it is the kind of thing that can be used to denote.

A representation X *of* Z denotes Z: Z must exist and X must denote Z.

But not all representations denote. A representation that does not denote can still represent something even if that something does not exist.

A picture of a fictional character represents the character, but is not a representation of the character because the character does not exist.

Goodman distinguishes between *representations-of-Z* and *Z-representations*, which can represent something that does not exist as a Z.

With this distinction at play we can define an initial notion of *representation-as*: X represents Y as Z when X is a Z-representation that denotes Y.

Additionally, we say that a representation *exemplifies* a property if it instantiates it and denotes it: a sample of some substance instantiates certain properties and (qua sample) represents them.

Hughes (1997)

These ideas inspire Hughes (1997) to provide an account of scientific representations that allows for surrogative reasoning.

The idea, roughly, is that such representations

1) *denote* their targets,
2) *demonstrate* certain properties (about themselves) and,
3) what these representations demonstrate is *interpreted* as being about the target.

Hence the view is called the DDI account.

Scientific representations are ‘about’ their targets in two senses: they denote them and are interpreted in terms of their targets.

Frigg summarizes: ‘a theory denotes its subject matter and a model denotes its target’ (MT, p. 275).

DEKI (Frigg and Nguyen)

According to the DEKI proposal, what makes something a Z-representation is that it is interpreted in a particular way: there has to be a key that, for every property of the object that serves as a representation that is deemed relevant, assigns a Z-property that is relevant (thus, for example, a unicorn-picture consists of an object that depicts a figure that has properties that can be keyed to unicorn-properties).

We can then characterize a Z-representation by a pair <O, I> where O is an object and I is a key of this sort.

We can then characterize a model as a kind of Z-representation, and say that a model M represents its target T as Z iff

1) M denotes T,
2) M exemplifies Z-properties P1…Pn,
3) M comes with a key associating P1…Pn with a set of properties Q1…Qn,
4) M imputes some of Q1…Qn to T.

Thus, similarly to how it was in Hughes' account, scientific representations are 'about' their targets in at least wo senses: they denote them and they are Z-representations that are interpreted in a terms that are suitable to talk about the targets.

Targetless models in DEKI and denotation

Obviously, DEKI can handle targetless models. Models are Z-representations, and for Z-representations it is not necessary that Z exists.

But what are they about?

- A heavy handed approach: they are about themselves. But then targetless models will have targets.
- More sophisticated: they are about the properties that they exemplify.
  - Elgin (2010) advances the idea that Z-representations constitute 'small genres' of all representations with a common-ostensible subject matter (p. 3).
  - Frigg dismisses the idea that models have genres.
  - Frig and Nguyen's DEKI also appeals to our capacity to select the relevant Z-properties.
- Since Z-representations exemplify Z-properties and key them up in different ways, we could say that they are about those properties. But exemplification entails denotation, so aboutness reduces into denotation.

The common source

Goodman's (1961) account of sentential aboutness. For X to be about Y, Y has to exist. Aboutness is selective: a sentence is about X only if it is not also about everything else. 'Everything is either red or not red' is about everything, but not particularly about me.

Goodman recognizes a secondary sense of 'being about', which he calls rhetorical aboutness.

- This tries to capture the distinction between the subject matter of 'Pegasus flew' and 'The Phoenix flew'. Since neither Pegasus not the Phoenix exist, there is nothing that they are both are absolutely about—so they are absolutely about the same subject matter (nothing).

Goodman's solution: introduce a notion of term-differential consequence, and then define a class of predicates of the form 'X-about' for sentences such that a sentence S is X-about if and only if there is some statement T that is a term-differential consequence of S with respect to the X-term. A sentence can be X-about without being absolutely about X.

This is effectively an early definition for Z-representations!

In order to say that some X is a Z-representation in this sense, it needs to be possible to define a notion of term-differential consequence with respect to X. It is not at all clear how this can be done when X is non-propositional, as in the case of pictures, sculptures, music… and (at least some) models.

This is a precursor of his later account of denotation.

Denotation and convention

Goodmanian's think that all denotation is conventional. Elgin (2010) is explicit:

> Representation-of can be achieved by fiat. We simply stipulate: let x represent y and x thereby becomes a representation of y. […] We could take any p-representation and stipulate that it represents any object. We might, for example, point to a tree-picture and stipulate that it denotes the philosophy department. But our arbitrary stipulation does not bring it about that the tree-representation represents the philosophy department as a tree. (p. 4)

Stipulation has limits concerning representation-as.

In Elgin's account, whether a Z-representation represents something as Z depends on whether it can be interpreted as a representation within a given genre. But this is to say that it has to be possible to identify its subject matter!

In Frigg's account, whether something represents-as-Z depends on whether there is an interpretation key that permits it to impute Z-properties to the target. Keys are assigned freely but not arbitrarily (cf. Frigg & Nguyen (2020), pp. 175–176, 181–182). Epistemic goals, background assumptions, and so on, all go into the selection of keys. While scientific representation has a stipulative dimension, it also is under constraints that go beyond the intention of scientists.

Frigg's account *has* to account for subject matters that are intentionally and non-intentionally determined.

Replacing denotation with aboutness

We take it that there is stock of subject matters.

A representation's subject matter S(R) is tracked by its users through an structure built out of of pairs <C, S>, where C is itself a subject matter, and S is a source.

Overall subject matter is the set of all the first elements of these pairs.

We suppose that subject matters hold relations to properties: then, by having a subject matter, a representation has relations to certain properties—we notate Z-properties as P(Z).

Objects induce subject matters. We notate: I(chess) = chess.

We say that subject matters can be included in, overlap, and so on, other subject matters.

R is a *Z-representation* if and only if it has Z as a subject-matter.

- A dog-representation is about dogs, a unicorn-representation is about unicorns.

Z-representations are related to Z-properties.

Two accounts

We get something like Levy's account:

Levy-like account A model of a target T is a Z-representation whose subject matter partially overlaps I(T).

Then, things that are true of M as a Z-representation are partially true of T.

We also get something like the DEKI account:

AKI A model represents a target T as Z iff

1) M is about Z,
2) M keys up features of Z (in particular, a relevant set of P(Z) that M talks about) to a set of properties G,
3) M imputes G to T.

The core idea of AKI is that what models say about their subject matter is interpreted in a way that allows the imputation of certain properties to the target. In this way, AKI retains the role of interpretation and imputation in DEKI, while it replaces the notions of denotation and exemplification with the framework concerning subject matters we have sketched.

AKI inherits many of the nice features of DEKI.

Unlike the Levy-like account, AKI does not assume that Z and I(T) overlap, so what M says about its subject matter need not be even partially true of T in order for M to be a model of the target.

Models and inquiry

Inquiry begins when a problem or question arises for some agent that can have an interest in addressing the problem or question.

This question can serve as a subject of inquiry.

This can become part of the intentionally determined subject matter of all representations involved in the process of inquiry.

Accordingly, the content of models subject-matter-wise includes inquisitive content: the questions at play.

Models contribute to inquiries by providing representations whose subject matter has promise in settling answers to the questions at play.

Successful models allow us to answer questions about their targets.

According to AKI, they do this by having subject matters which can be interpreted as telling about properties of the targets.

This is similar to Knuutilla's (2011, 2021) idea that models are tools for question-solving, and Morgan and Morrison's (1999) view of models as autonomous agents that serve as instruments of investigation.

- However, in AKI models are not representations-for question-oriented-tasks, but also representations-of questions.

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¡Gracias!



fmoralesc.github.io



ef.em.carbonell@gmail.com



okf@scholar.social